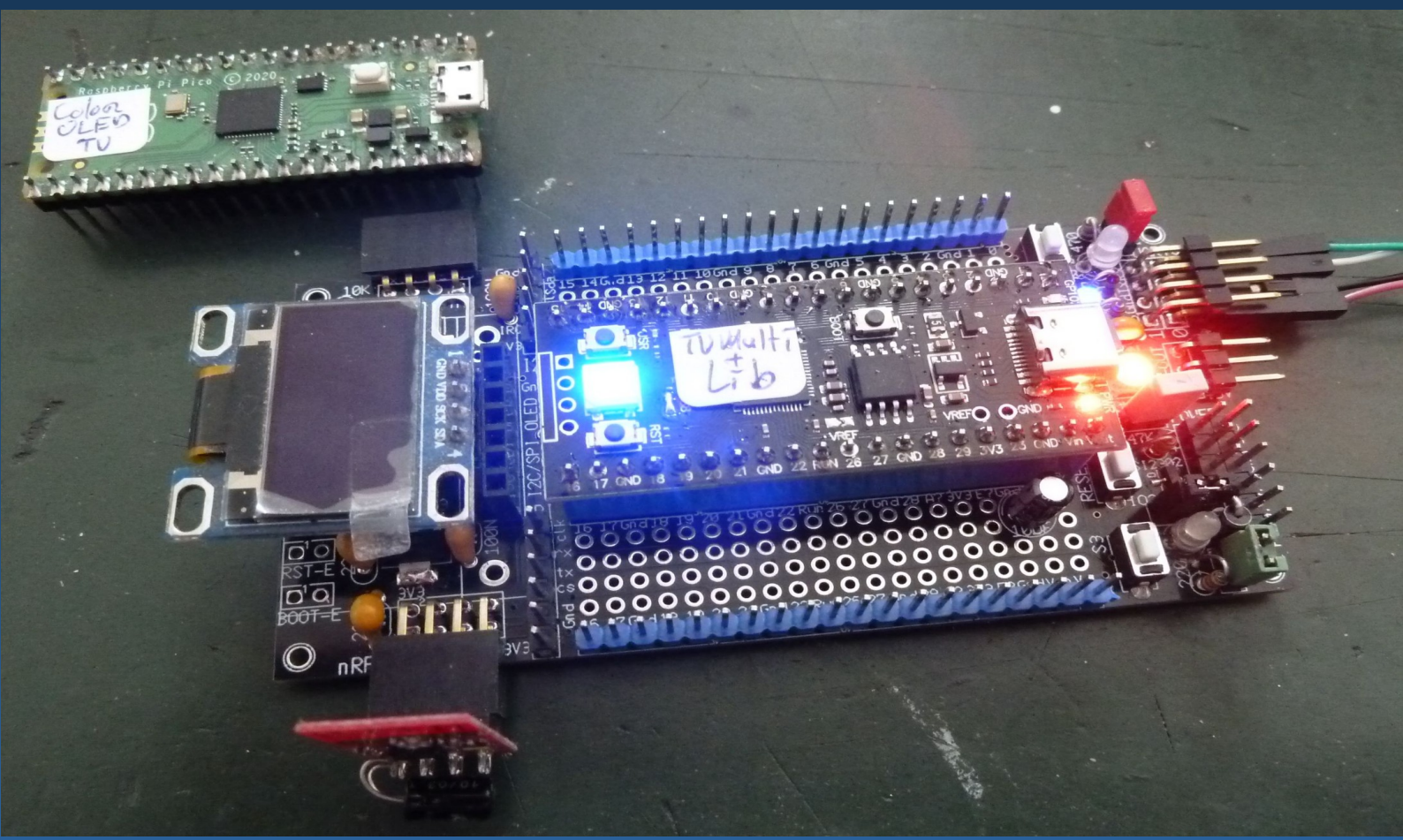
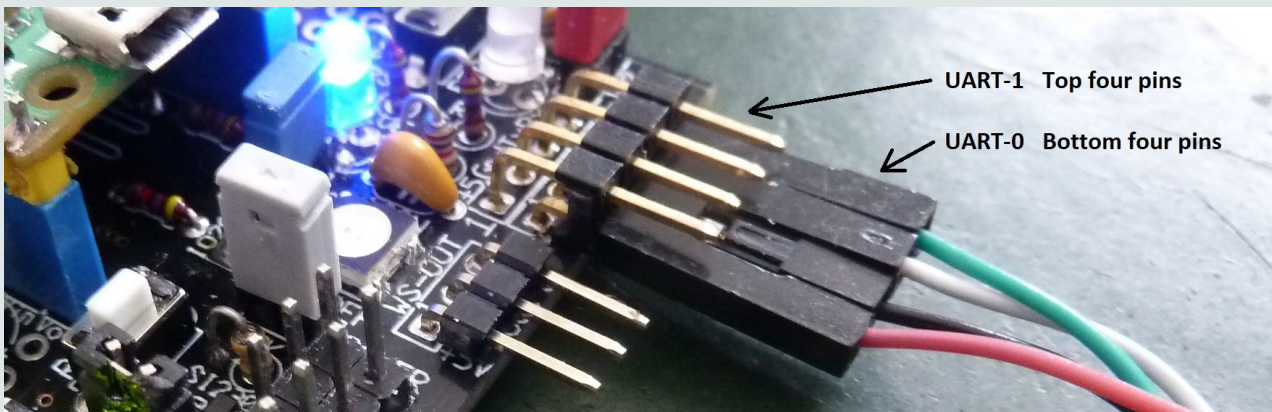


Pico-kit v2a manual

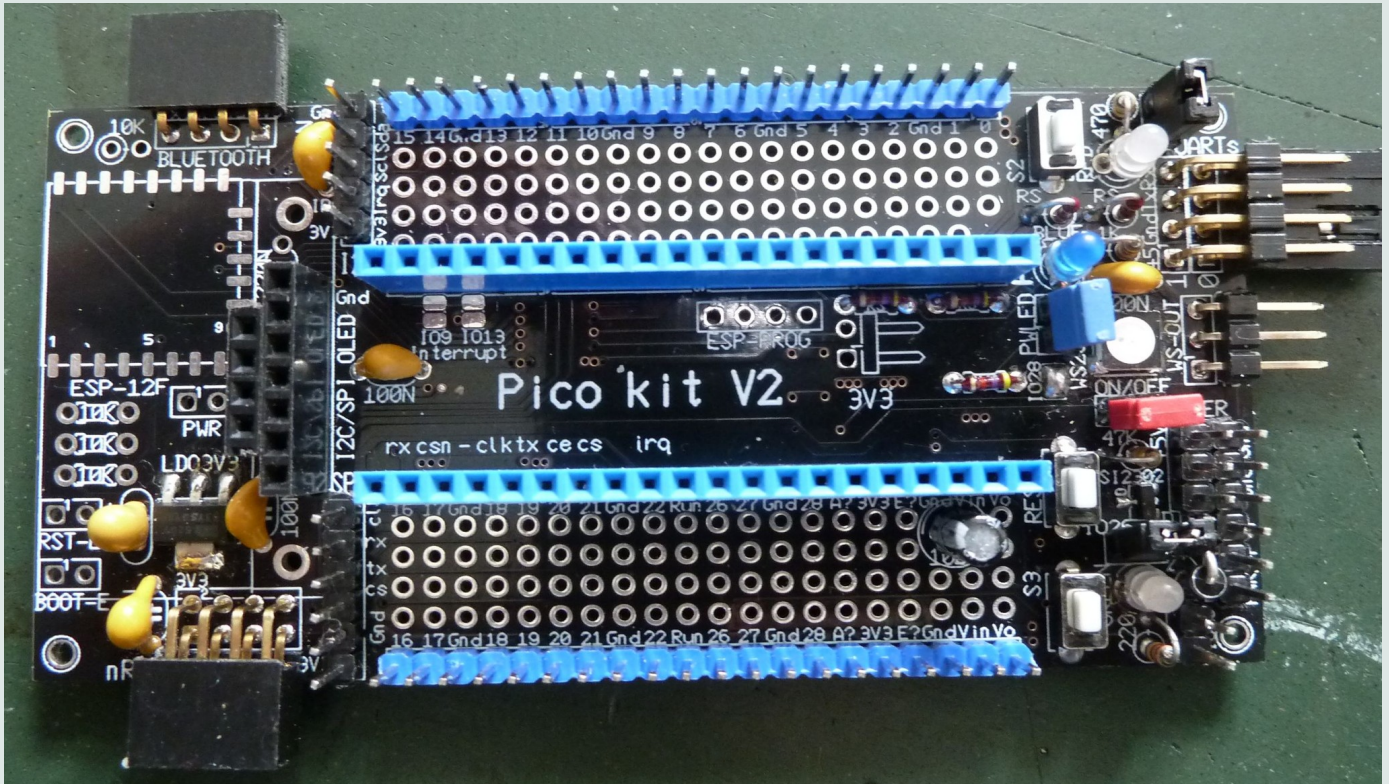


Pico-kit I/O

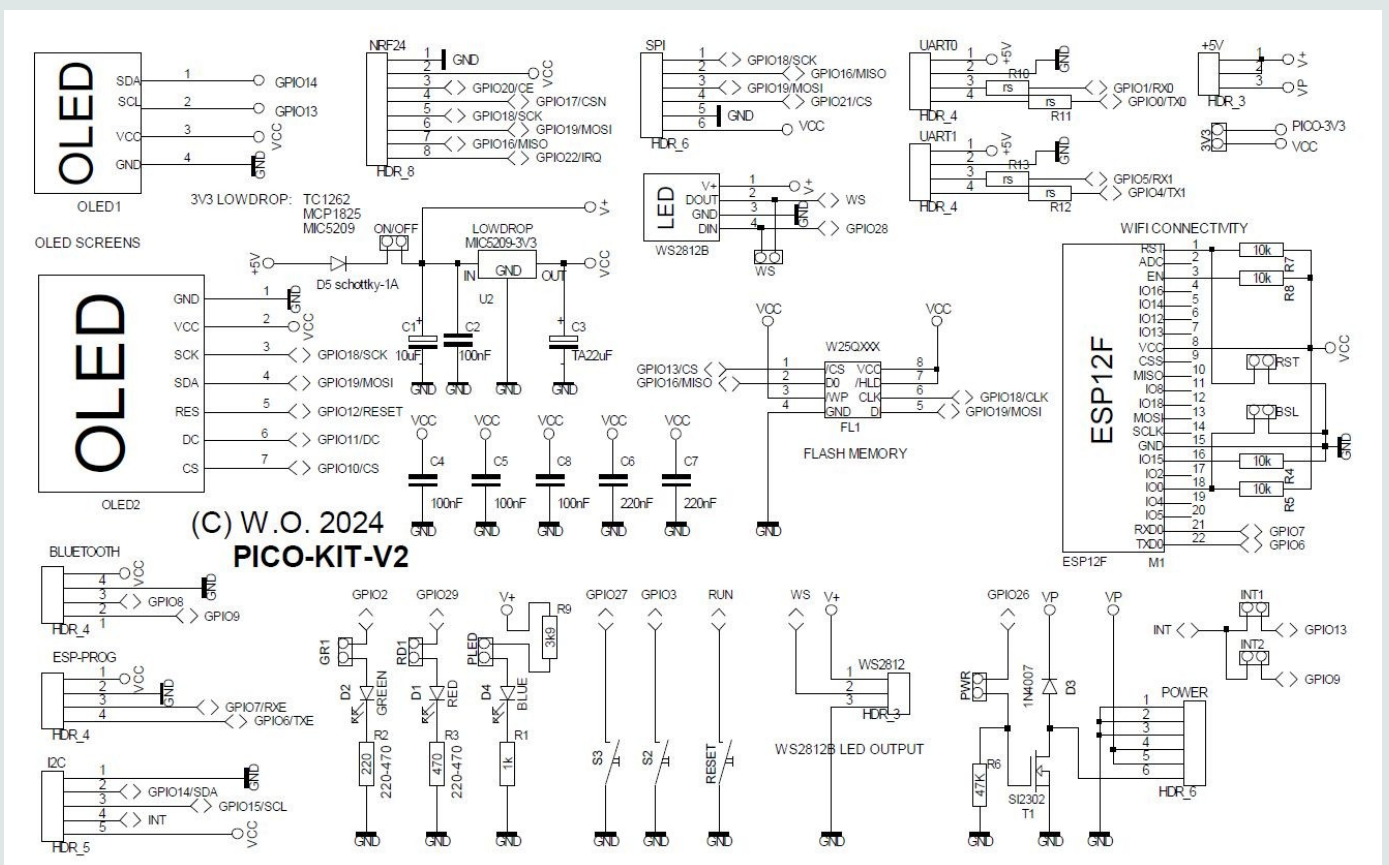
GPIO	Function on Pico-kit and CPU boards
0	RS232 TX0 (UART-0)
1	RS232 RX0
2	Red LED
3	Switch S2
4	RS232 TX1 (UART01)
5	RS232 RX1
6	ESP-12F TX
7	ESP-12F RX
8	Bluetooth TX
9	Bluetooth RX/I ² C interrupt input
10	SPI OLED CS
11	SPI OLED DC
12	SPI OLED RES(et)
13	I ² C interrupt input or SPI Flash /CS
14	I ² C SDA
15	I ² C SCL
16	SPI0 Miso
17	NRF24L01 CSN
18	SPI0 Clk
19	SPI0 Mosi
20	nRF24L01 CE
21	SPI header CS
22	NRF24L01 IRQ
23	WS2812 (on YD-RP2040) or Power save (on Pico board)
24	USR switch (on YD-RP2040) or VBUS ADC in (on Pico board)
25	LED on Pico (compatible) board
26	Power out to MOSFET
27	Switch S3
28	WS2812
29	VSYS ADC in (on Pico board) or green LED



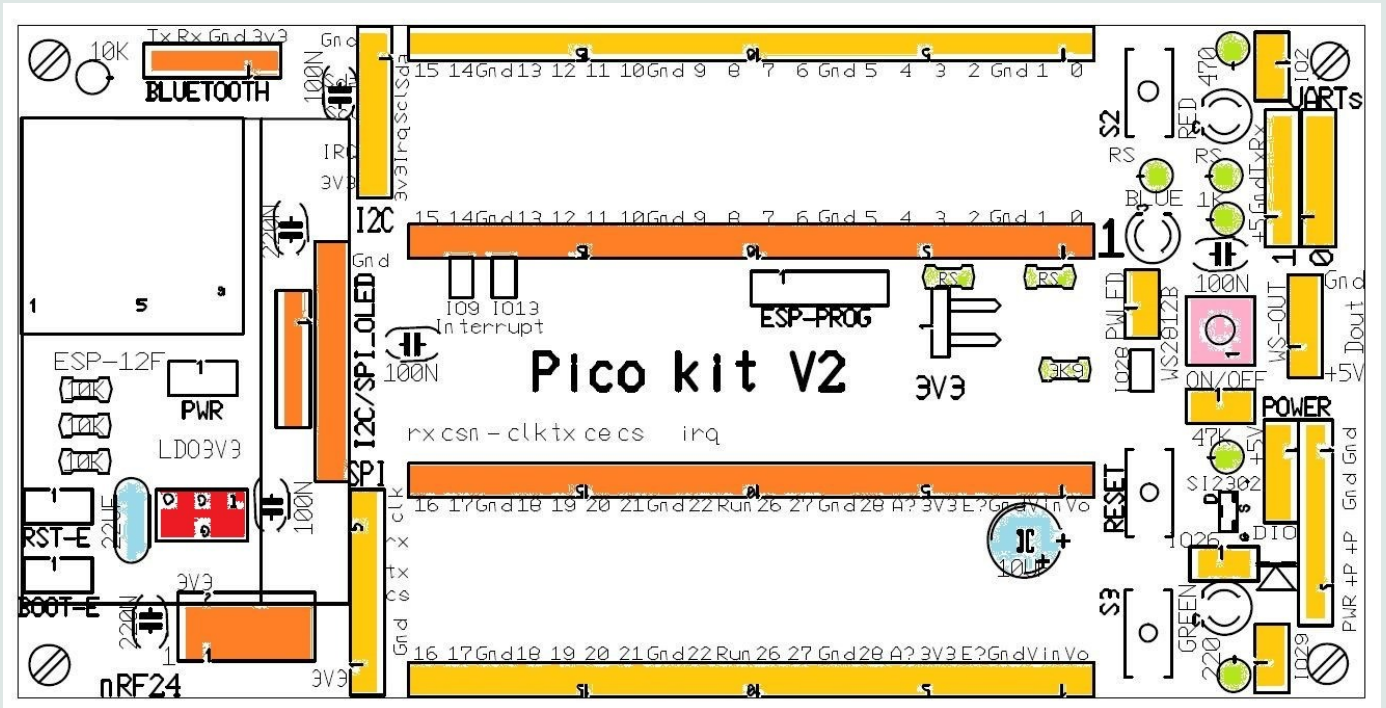
Constructing the Pico-kit



Pico-kit v2 schematics

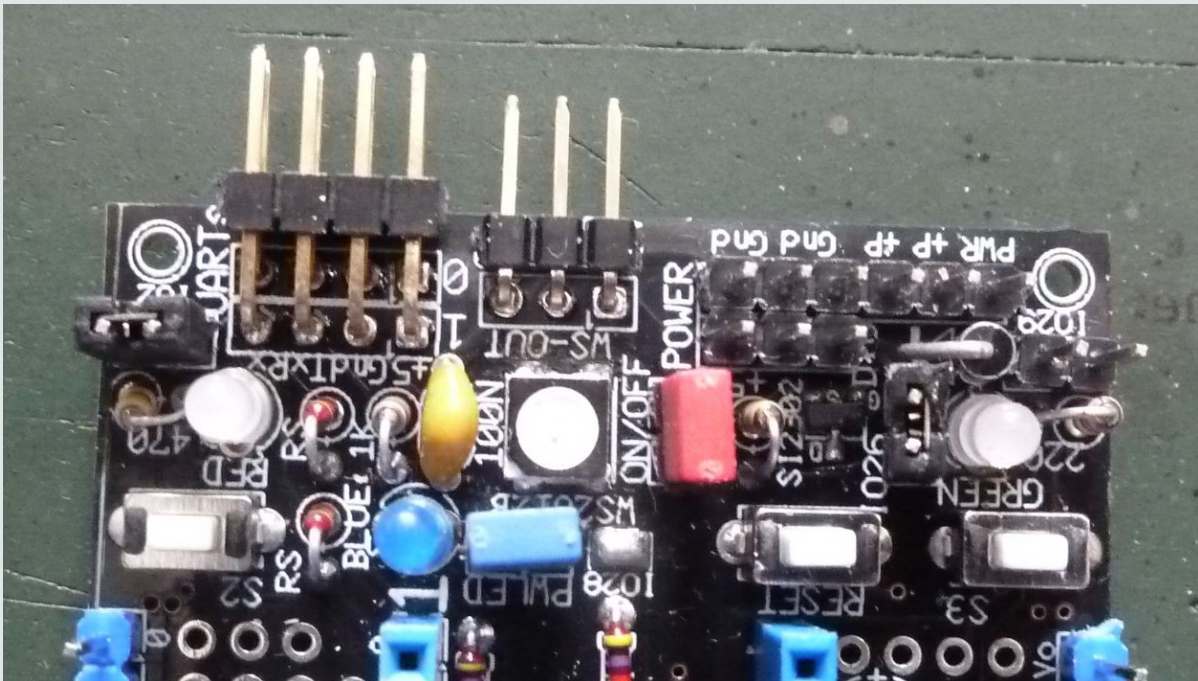


Parts placing



- Green = Resistors
- Pink = WS2812
- Red = MIC5209
- Blue = Electrolytic capacitors
- Yellow = Male headers
- Orange = Female headers

Pico-kit front end

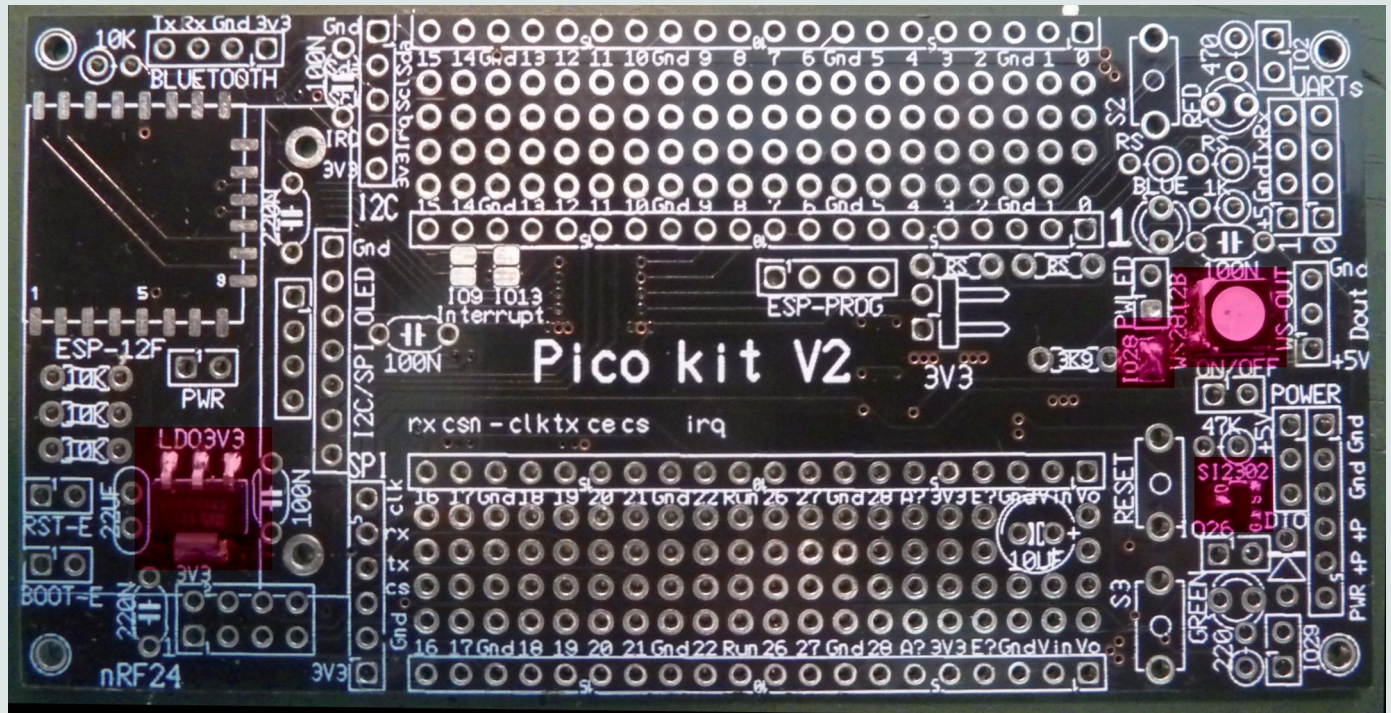


Construction of the Pico-kit

Step-1

1) First the **red** solder bridge, then the SMD parts, the MOSFET, WS2812 & MIC5209 and optional ESP-12F

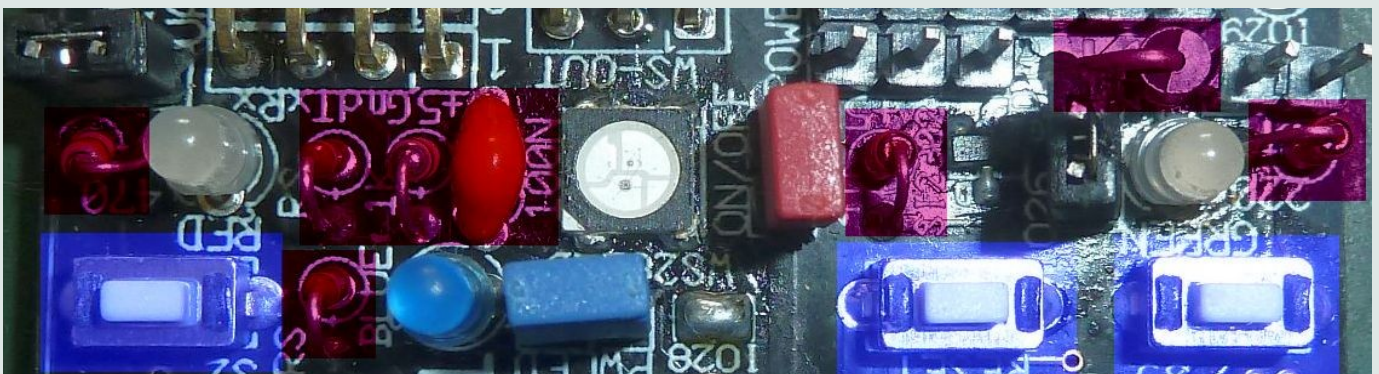
1a) Turn the board upside down and mount the SMD schottky-diode on the bottom side of the board (the position is marked SS34)



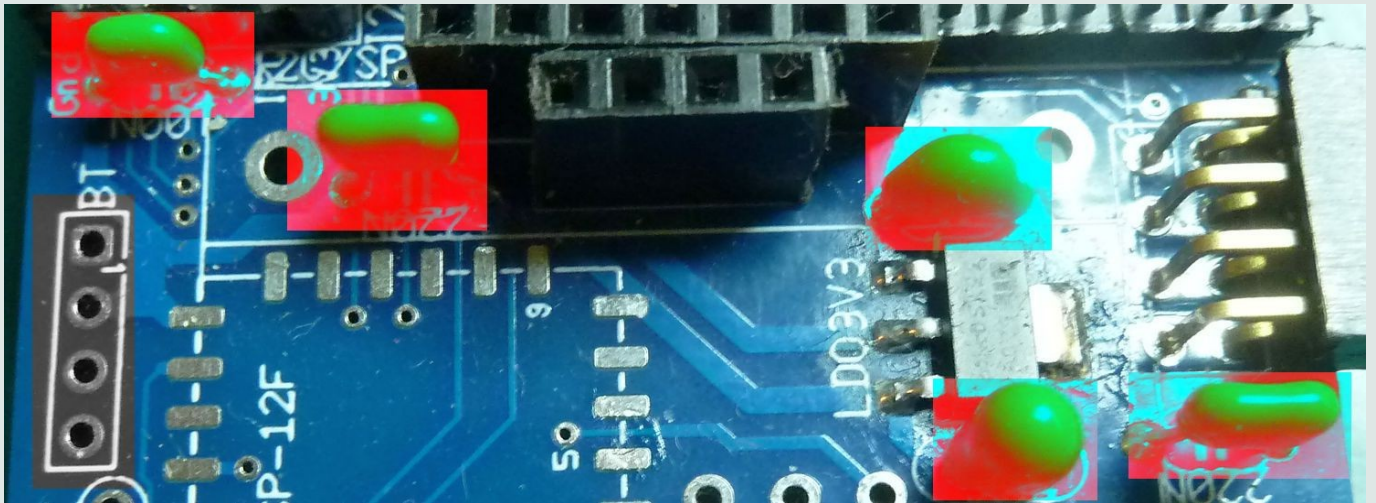
Step-2

2) We do not use the 3V3 jumper & ESP programming connector!

3a) Then the three switches all resistors, capacitors and the diode. The switches – **blue**, other parts **red**

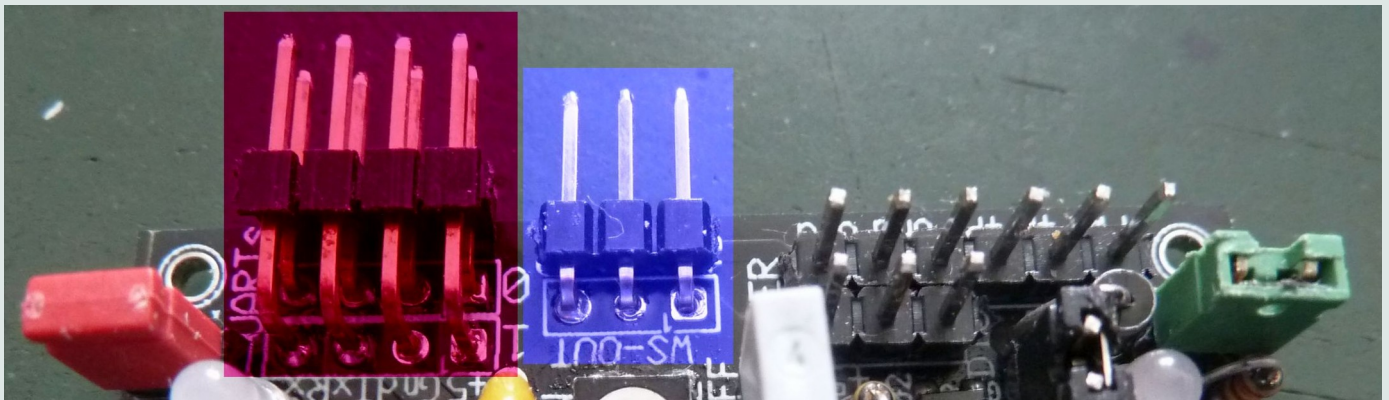


3b) Some more capacitors in **red** and then the right-angle header for the nRF24

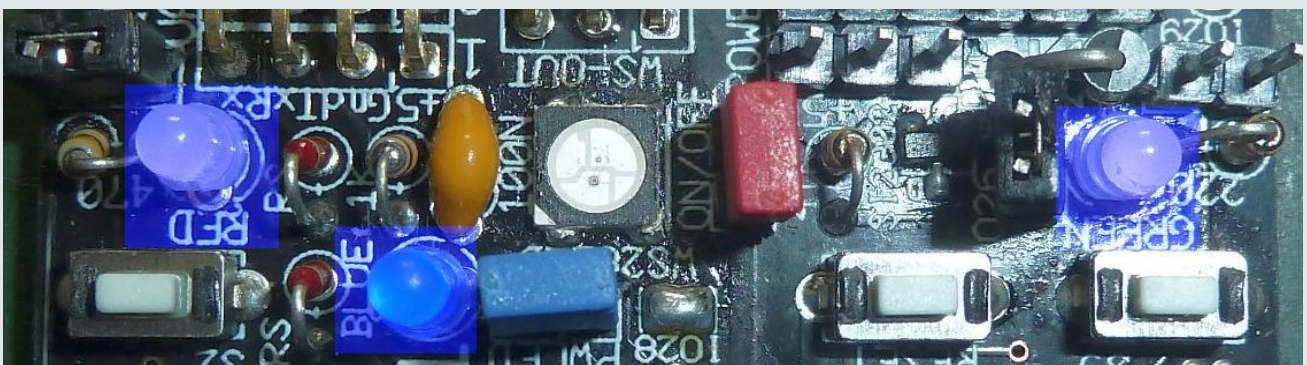


Step-3

4) Now the **blue** 3-pins squared header, then the **red** 8-pins dual row squared header

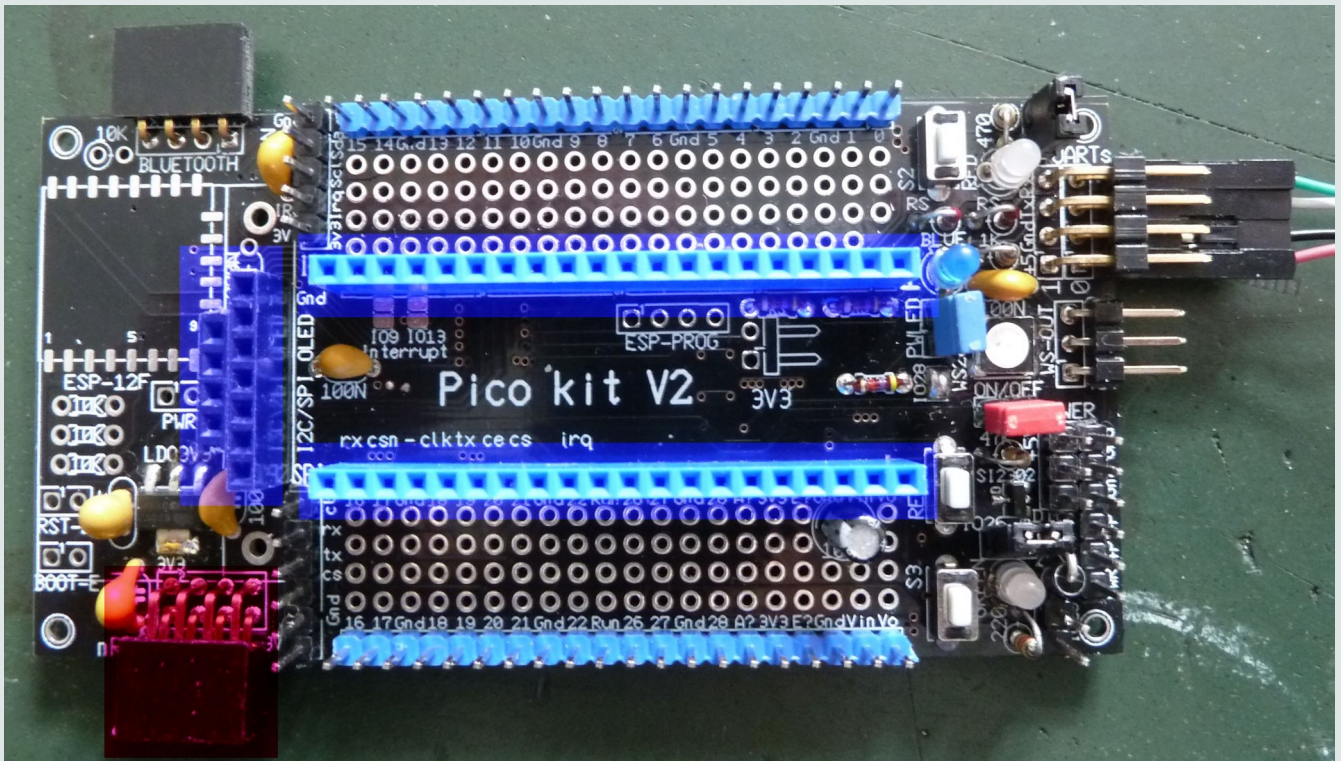


5) The three LEDs, not too high otherwise the remaining headers will be difficult to place. Note, the not colored leds have the long wire and flat spot at the lower position in this picture!

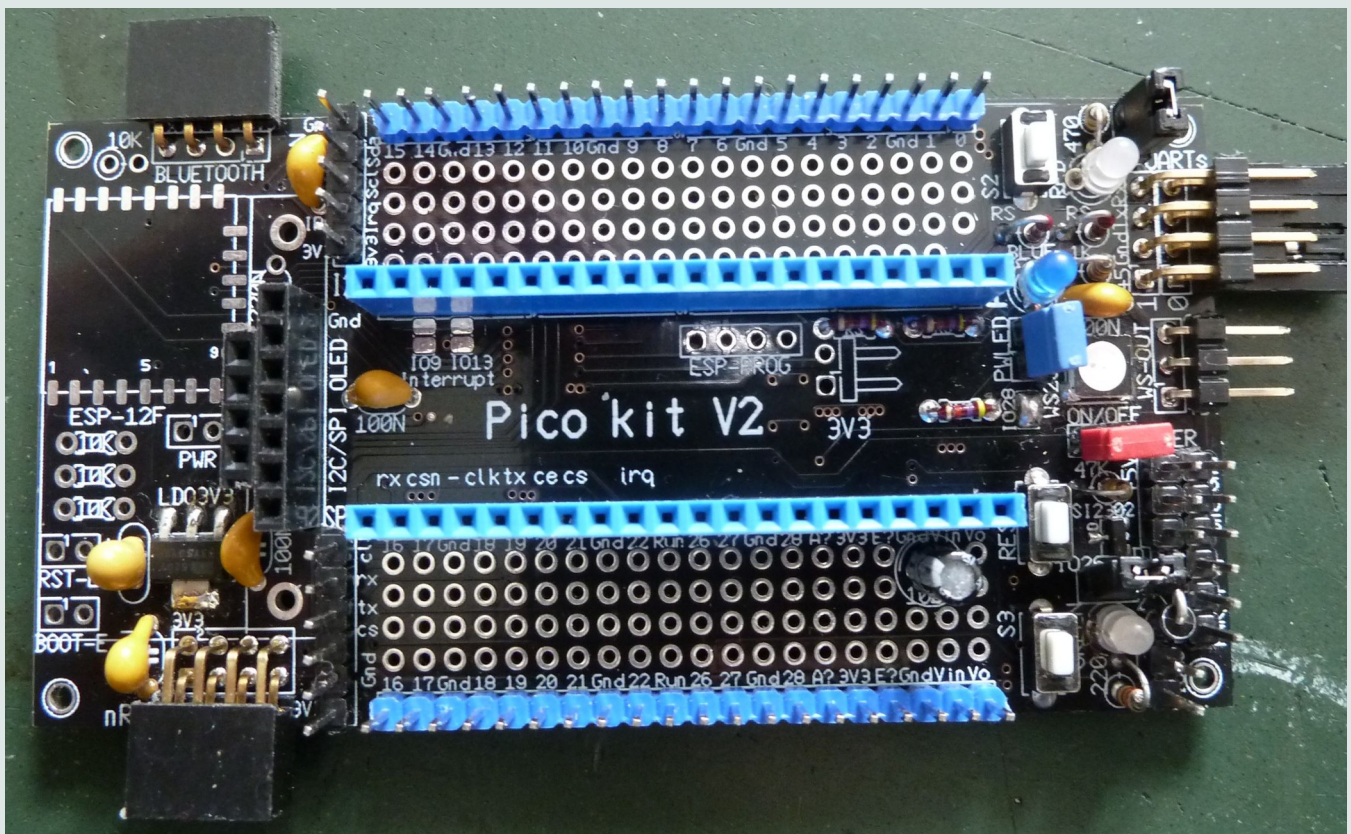


Finish

- 6) The female headers first, then the male headers, place them carefully vertically! NOTE the not present optional ESP-12F!



Almost completed



Bill of present materials

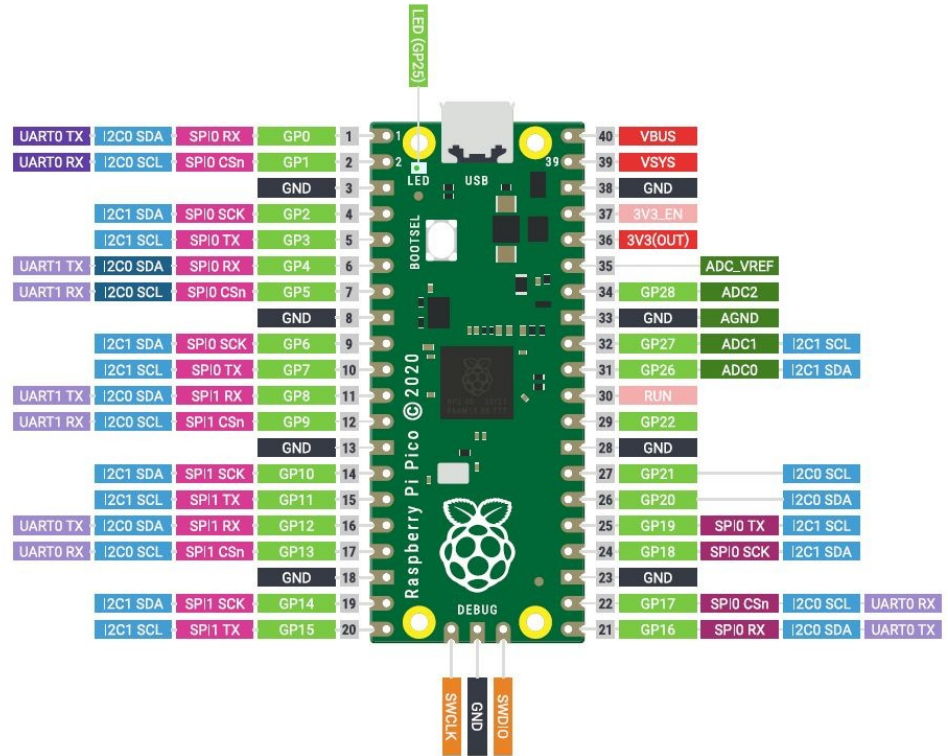
Number	Component
1	PCB board
4	RS = 470 Ω
1	220 Ω
1	470 Ω
1	1 k Ω & 4k7
1	47 k Ω
3	100nF
1	220nF
1	22 μ F tant.
1	10 μ F elco
2	3mm LED gr/rd
1	3mm LED blue
1	1N4007, SS34 (SMD Schottky)
1	Mosfet SI2302
1	WS2812B LED
1	MIC5209-3.3ys
3	Switch
5	2 pins male header
1	3 pins male header & 3 pins squared male header
1	2x4 pins squared male header
1	5 pins male header
2	6 pins male header
2	20 pins male header
1	4 pins female header
1	7 pins female header
2	20 pins female header
1	2x4 pins square female header
5	jumper
4	Female jumper wire

Bill of optional parts (not present)

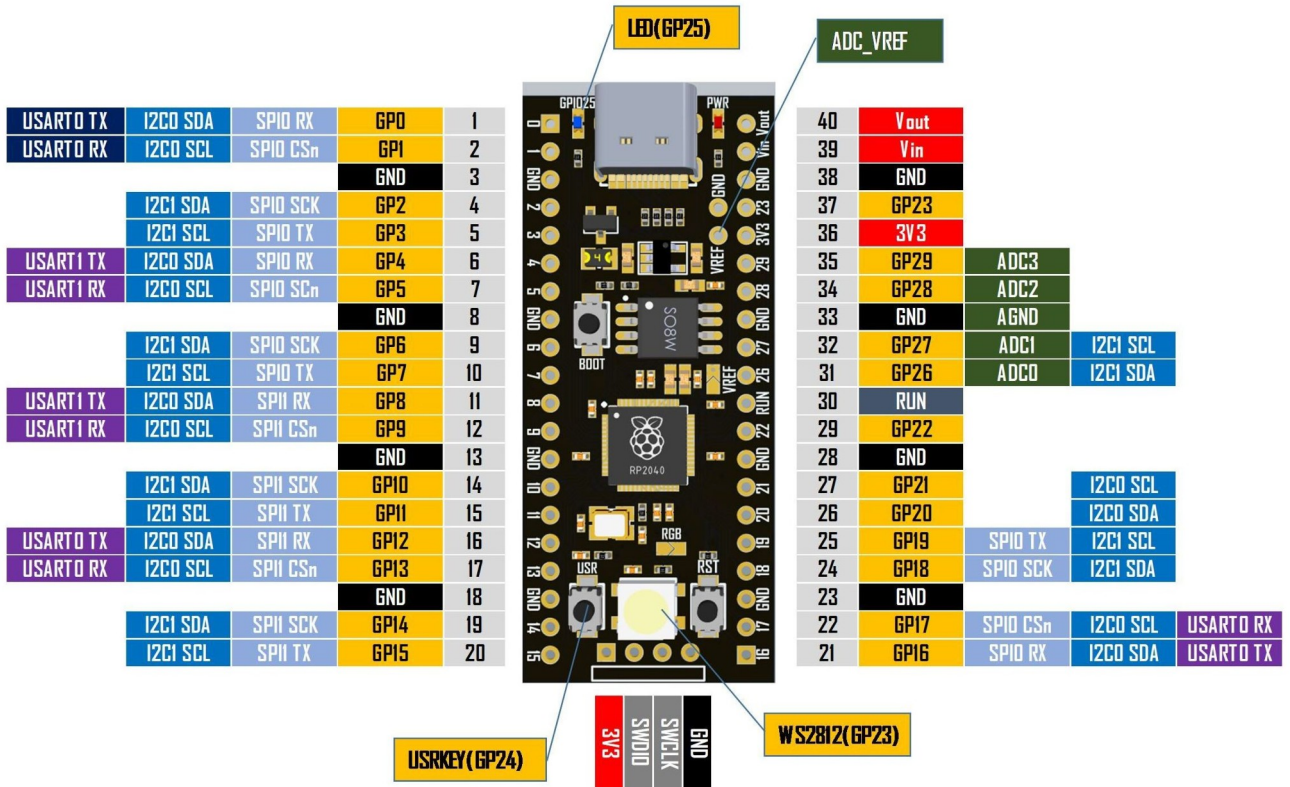
1	220nF
1	47k Ω
4	10 k Ω
3	2 pins male header
1	4 pins male header
1	4 pins squared female header
3	jumper
1	2 pins squared male header
1	W25Q16BV to W25Q128BV
1	Bluetooth module (HC-05/HC-06)
1	ESP-12F wifi module
1	SSD1306 or SSD1351 OLED with I ² C or SPI bus
1	Pi Pico board or YD-RP2040 or RP2040 Plus, etc.
1	nRF24L01P 2.4GHz transceiver

Original Pico board

Power
Ground
UART / UART (default)
GPIO, PIO, and PWM
ADC
SPI / SPI (default)
I2C / I2C (default)
System Control
Debugging

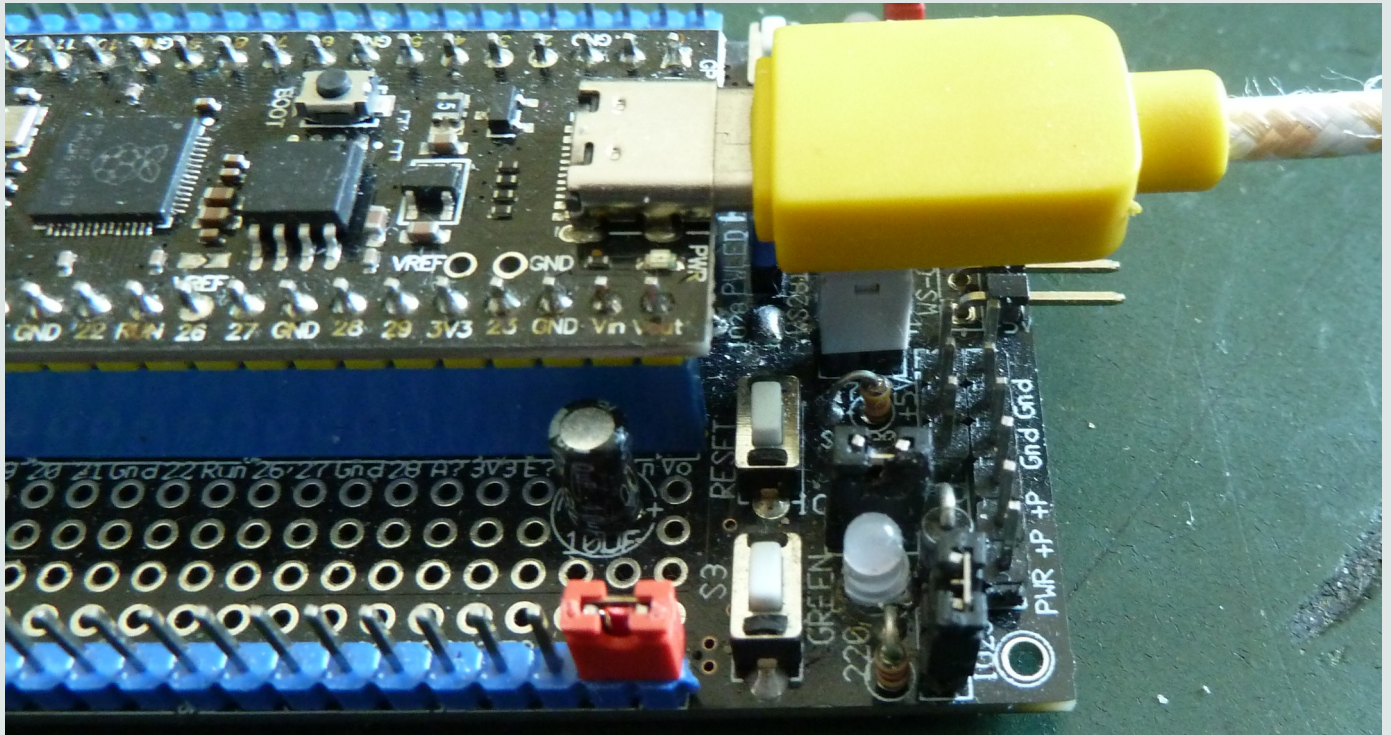


(VCC-GND) YD-RP2040 board



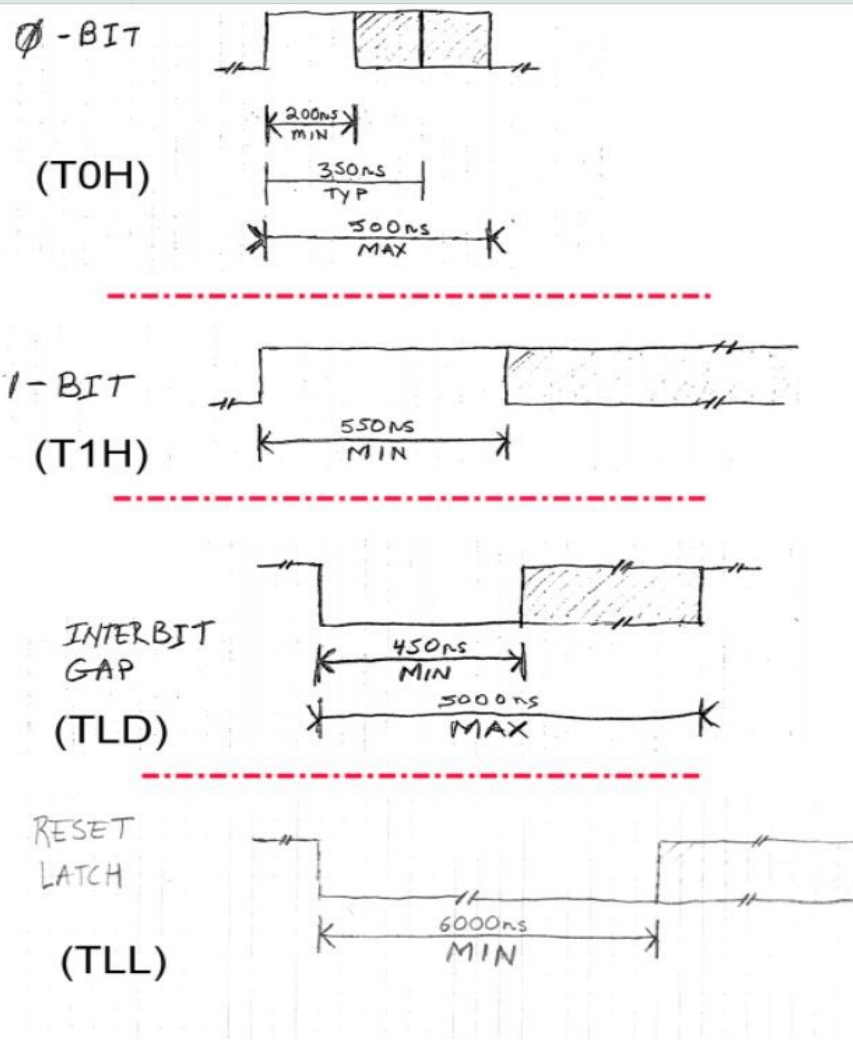
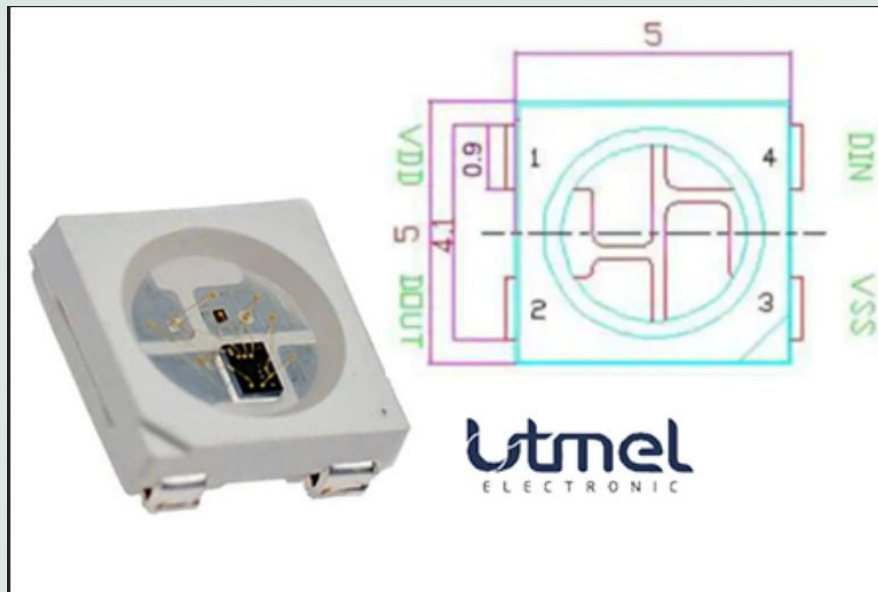
USB adaptation

When using a USB-cable for communication with the Pico-kit, you need to place one extra jumper on the pins $V_{in} - V_o$. As shown on the picture below. This jumper connects the USB-power to the 5 Volt circuit on the Pico-kit board.



Note the red jumper next to switch S3

WS2812B a full color digitally controlled serial LED



WS2812B control signals

nRF24L01P transceiver module

1. Introduction

E01-ML01D



E01-ML01D is a 1mW , industrial-grade wireless transceiver module, operates at 2.4GHz with high air data rate (2Mbps maximum). SPI interface, inline pin, high stability, cost-effective and Batch production, which makes the module is suitable for various applications.

E01-ML01D is based on original imported nRF24L01P form Nordic in Norway. The module uses industrial component with sufficient transmitting power, good spectral characteristics and small volumes. With high-performance PCB antenna and impedance matching precision, there is no need for an external antenna for embedded development.

2. Electrical parameter

E01-ML01D

No.	Parameter item	Parameter details	Description
1	RF IC	nRF24L01P	Nordic
2	Size	12.6 * 22.6 mm	
3	Production process	Machine	Lead-free
4	Connector	2 * 4 * 2.54mm	Plug-in
6	Supply voltage	1.9 ~ 3.6V DC	Notes: the voltage higher than 3.6V is forbidden
3	Frequency	2400 ~ 2525MHz	Adjustable
7	Communication level	0.7VCC ~ 3.6V	VCC refers to the supply voltage
8	Operation Range	100m	Test condition: Clear and open area, 0dBm , antenna gain: 5dBi , height: 2m , Air date rate: 250Kbps
9	Max Power	Maximum 0dbm	About 1mW
10	Air data rate	3 level adjustable	250kbps、1Mbps、2Mbps
11	Sleep current	1.0uA	nRF24L01P sets as power-down
12	Transmitting current	13mA@0dBm	The largest emission current
13	Receiving current	13mA	CE=1
14	Communication interface	SPI	Data rate: up to 10Mbps
15	Transmitting length	3 level FIFO.	32 bytes (maximum) for one package
16	Receiving length	3 level FIFO.	32 bytes (maximum) for one package
17	RSSI support	N/A	Support simple data packet loss statistics
18	Antenna type	PCB	50 ohm characteristic impedance
19	Sensitivity	-106dBm	250kbps
20	Operating temperature	-40 ~ +85℃	Industrial-grade
21	Operating humidity	10% ~ 90%	Relative humidity, without condensation
22	Storage temperature	-40 ~ +125℃	Industrial-grade

HC05 to HC08 Bluetooth module

HC-08 BLUETOOTH UART COMMUNICATION MODULE USER MANUAL

Version

Software:HC-08 V3.1, Hardware: V2.0

Date

2017-07-07

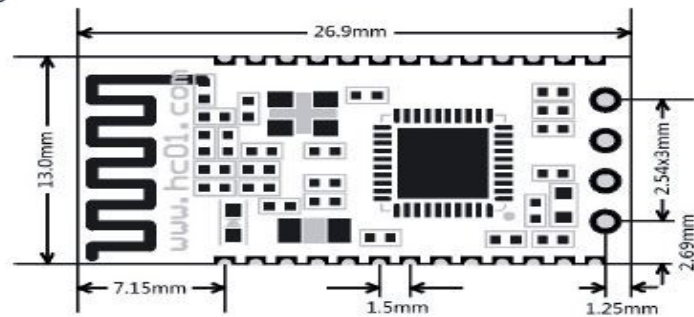
Product introduction

HC-08 Bluetooth UART communication module is a new generation of Bluetooth specification V4.0 BLE Bluetooth protocol based on the transmission module. Wireless working frequency is 2.4GHz ISM, modulation is GFSK. The maximum transmit power module 4dBm, the receiving sensitivity is -93dBm, and iphone4s can achieve 80 meters of super long distance communication under open environment.

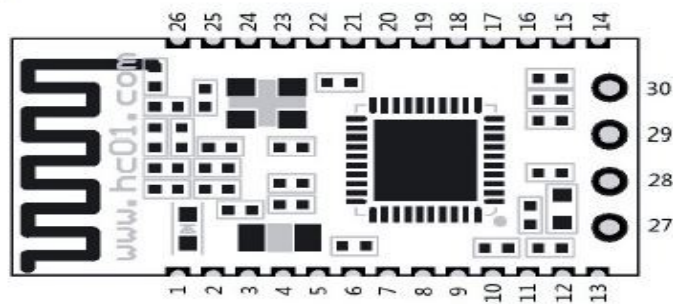
The module uses the stamp hole encapsulation, can patch welding, module size is 26.9mm * 13mm * 2.2mm, very convenient to the customer within the embedded application system.

The module uses the CC2540 chip, the configuration of the 256K Byte space, supports AT command, the user can according to need to change role and the serial baud rate, equipment name and other parameters, the use of flexible.

Product size



Pin definition



WEB: www.hc01.com

PAGE 1

W25Q16 to W25Q128 Flash memory

3. PACKAGE TYPES AND PIN CONFIGURATIONS

3.1 Pin Configuration SOIC 150/208-mil

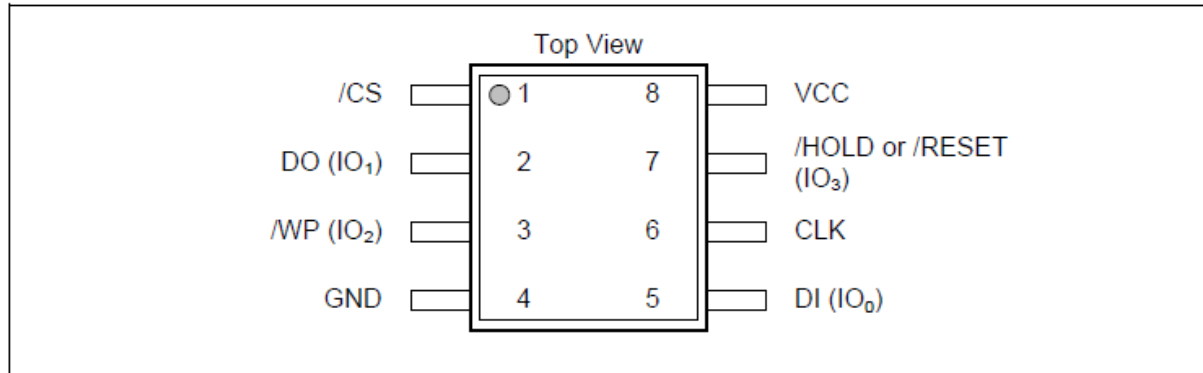


Figure 1a. Pin Assignments, 8-pin SOIC150-mil/208-mil (Package Code SN, SS)

Description

The W25Q64BV (64M-bit) Serial Flash memory provides a storage solution for systems with limited space, pins and power. The 25Q series offers flexibility and performance well beyond ordinary Serial Flash devices. They are ideal for code shadowing to RAM, executing code directly from Dual/Quad SPI (XIP) and storing voice, text and data. The devices operate on a single 2.7V to 3.6V power supply with current consumption as low as 4mA active and 1µA for power-down. All devices are offered in space-saving packages.

The W25Q64BV array is organized into 32,768 programmable pages of 256-bytes each. Up to 256 bytes can be programmed at a time. Pages can be erased in groups of 16 (sector erase), groups of 128 (32KB block erase), groups of 256 (64KB block erase) or the entire chip (chip erase). The W25Q64BV has 2,048 erasable sectors and 128 erasable blocks respectively. The small 4KB sectors allow for greater flexibility in applications that require data and parameter storage. (See figure 2.)

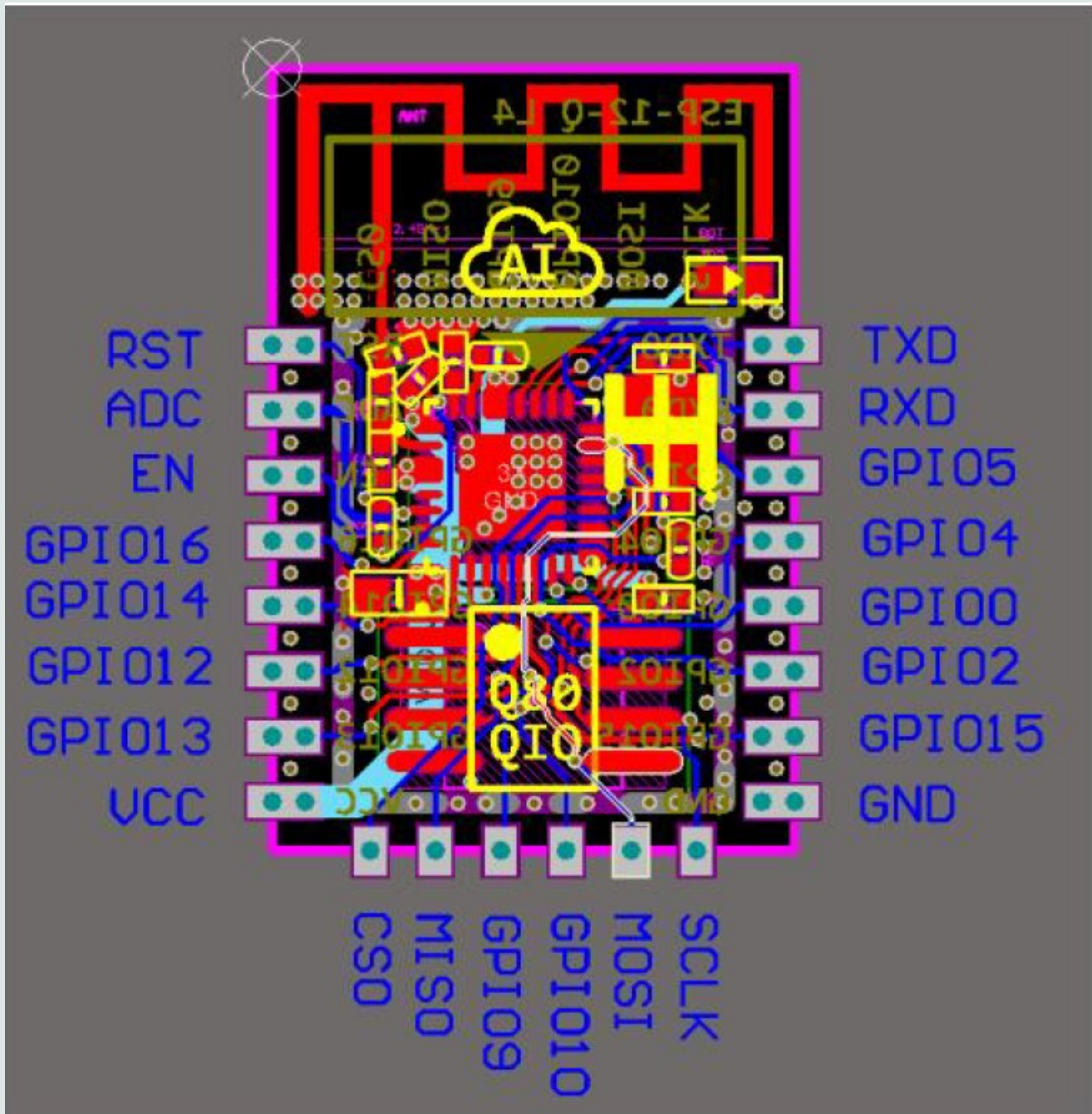
The W25Q64BV supports the standard Serial Peripheral Interface (SPI), and a high performance Dual/Quad output as well as Dual/Quad I/O SPI: Serial Clock, Chip Select, Serial Data I/O0 (DI), I/O1 (DO), I/O2 (/WP), and I/O3 (/HOLD). SPI clock frequencies of up to 80MHz are supported allowing equivalent clock rates of 160MHz for Dual Output and 320MHz for Quad Output when using the Fast Read Dual/Quad Output instructions. These transfer rates can outperform standard Asynchronous 8 and 16-bit Parallel Flash memories. The Continuous Read Mode allows for efficient memory access with as few as 8-clocks of instruction-overhead to read a 24-bit address, allowing true XIP (execute in place) operation.

Features

- **Family of SpiFlash Memories**
 - W25Q64BV: 64M-bit / 8M-byte (8,388,608)
 - 256-bytes per programmable page
- **Standard, Dual or Quad SPI**
 - Standard SPI: CLK, /CS, DI, DO, /WP, /Hold
 - Dual SPI: CLK, /CS, IO₀, IO₁, /WP, /Hold
 - Quad SPI: CLK, /CS, IO₀, IO₁, IO₂, IO₃
- **Highest Performance Serial Flash**
 - Up to 6X that of ordinary Serial Flash
 - 80MHz clock operation
- **Low Power, Wide Temperature Range**
 - Single 2.7 to 3.6V supply
 - 4mA active current, <1µA Power-down (typ.)
 - -40°C to +85°C operating range
- **Flexible Architecture with 4KB sectors**
 - Uniform Sector Erase (4K-bytes)
 - Block Erase (32K and 64K-bytes)
 - Program one to 256 bytes
 - More than 100,000 erase/write cycles
 - More than 20-year data retention

ESP12F

- 2.4 GHz Transceiver 802.11 b/g/n
- Low power 32-bit MCU
- 10-bit ADC
- TCP/IP protocol stack
- Deep sleep < 10 μ A
- Wakeup within 2 millisec.
- I/O, UART, I2C, PWM



Si2302 MOSFET

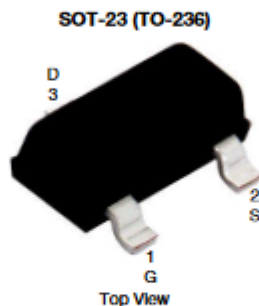


www.vishay.com

Si2302CDS

Vishay Siliconix

N-Channel 20 V (D-S) MOSFET



Marking code: N2

PRODUCT SUMMARY

V_{DS} (V)	20
$R_{DS(on)}$ max. (Ω) at $V_{GS} = 4.5$ V	0.057
$R_{DS(on)}$ max. (Ω) at $V_{GS} = 2.5$ V	0.075
Q_g typ. (nC)	3.5
I_D (A)	2.9
Configuration	Single

FEATURES

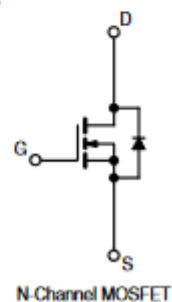
- TrenchFET® power MOSFET
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

APPLICATIONS

- Load switching for portable devices
- DC/DC converter



RoHS
COMPLIANT
HALOGEN
FREE
Available



ORDERING INFORMATION

Package	SOT-23
Lead (Pb)-free	Si2302CDS-T1-E3
Lead (Pb)-free and halogen-free	Si2302CDS-T1-GE3

ABSOLUTE MAXIMUM RATINGS ($T_A = 25$ °C, unless otherwise noted)

PARAMETER	SYMBOL	5 S	STEADY STATE	UNIT
Drain-source voltage	V_{DS}	20	20	V
Gate-source voltage	V_{GS}	± 8	± 8	V
Continuous drain current ($T_J = 150$ °C) ^a	I_D	2.9	2.6	A
		2.3	2.1	
Pulsed drain current ^b	I_{DM}	10	10	A
Continuous source current (diode conduction) ^a	I_S	0.72	0.6	A
Power dissipation ^a	P_D	0.86	0.71	W
		0.55	0.46	
Operating junction and storage temperature range	T_J, T_{stg}	-55 to +150	-55 to +150	°C

THERMAL RESISTANCE RATINGS

PARAMETER	SYMBOL	TYPICAL	MAXIMUM	UNIT
Maximum junction-to-ambient ^a	R_{thJA}	120	145	°C/W
		140	175	
Maximum junction-to-foot	R_{thJF}	62	78	°C/W

Notes

- a. Surface mounted on 1" x 1" FR4 board
b. Pulse width limited by maximum junction temperature



MIC5209

500 mA Low-Noise LDO Regulator

Features

- Output Voltage Range: 1.8V – 15V
- Meets Intel® Slot 1 and Slot 2 Requirements
- Guaranteed 500 mA Output Over the Full Operating Temperature Range
- Low 500 mV Maximum Dropout Voltage at Full Load
- Extremely Tight Load and Line Regulation
- Thermally Efficient Surface-Mount Package
- Low Temperature Coefficient
- Current and Thermal Limiting
- Reversed-Battery Protection
- No-Load Stability
- 1% Output Accuracy
- Ultra-Low-Noise Capability in SOIC-8 and DPAK versions
- Ultra-Small 3 mm × 3 mm DFN Package

Applications

- Pentium II Slot 1 and Slot 2 Support Circuits
- Laptop, Notebook, and Palmtop Computers
- Cellular Telephones
- Consumer and Personal Electronics
- SMPS Post-Regulator and DC/DC Modules
- High-Efficiency Linear Power Supplies

General Description

The MIC5209 is an efficient linear voltage regulator with very low dropout voltage, typically 10 mV at light loads and less than 500 mV at full load, with better than 1% output voltage accuracy.

Designed especially for hand-held, battery-powered devices, the MIC5209 features low ground current to help prolong battery life. An enable/shutdown pin on the SOIC-8 and DPAK versions can further improve battery life with near-zero shutdown current.

Key features include reversed-battery protection, current limiting, overtemperature shutdown, ultra-low-noise capability (SOIC-8 and DPAK versions), and is available in thermally efficient packaging. The MIC5209 is available in adjustable or fixed output voltages.

Typical Application Circuits

